Probabilistic Machine Learning

Lecture 04: Linear Models — From Least Squares to Logistic Regression

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Speaker notes

One session. The students know linear and logistic regression; the lecture reads them as probability models. Module 1 is the classical account of linear regression as taught earlier (closed form, R^2, basis functions, LMS, Gauss–Markov) — the shared Data Science units, unchanged. Module 2 asks what squared error silently assumed and shows that it is a Gaussian likelihood (keep the peak, Lecture 03, applied to a regression). Module 3 swaps the Gaussian for a Bernoulli: logistic regression, and shortcut 3 — model only $p(y \mid x)$. Module 4 sets it against naive Bayes of Lecture 03 (the same linear log-odds, fitted two ways). Module 5 answers Lecture 01's open question of when a score is a probability. Running examples: house prices for flats of 40–120 m^2 and the 600 m^2 villa of Lecture 01 (regression), and fraud with a base rate of 1 in 1,000 (classification).

2026-10-06: swapped with the former Lecture 04. The classical account moved from the backup into Module 1; the one-slide regression recap was dropped.

Where We Left Off

Lecture 03 built a classifier from probability and took two shortcuts on the way:

Shortcut	Instead of the ideal	What is dropped
Keep the peak	the most probable parameter (MLE, MAP)	the spread: how sure we are
Assume it apart	factorise the joint — naive Bayes	the dependences between attributes

Today we meet the methods you know from earlier courses — linear and logistic regression — and read them as **probability models**. They never model the inputs:

$$p(y \mid x) \quad \text{instead of} \quad p(x, y) = p(y) \cdot p(x \mid y)$$

SHORTCUT 3

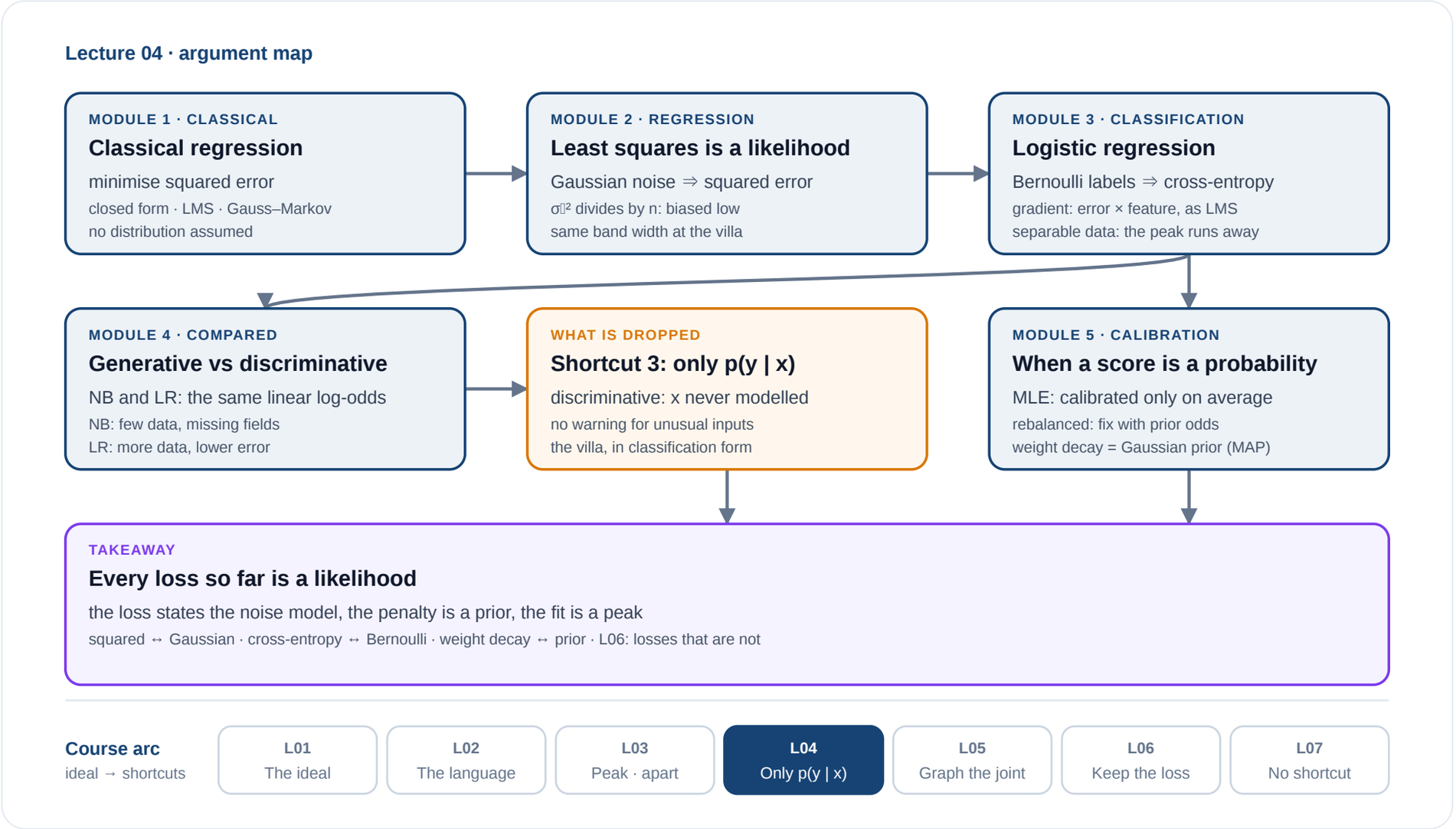
Only what you need: model the label given the inputs — *discriminative*.

Speaker notes

Naive Bayes needed the whole joint, so it paid with an independence assumption. Today's models take the other way out: skip $p(x)$ altogether. Plan: Module 1 is classical linear regression exactly as the students know it (it was the backup of the earlier version of this lecture); Module 2 asks what squared error assumed; Module 3 turns the same idea into logistic regression; Module 4 sets logistic regression against naive Bayes, which turn out to share the linear log-odds; Module 5 answers Lecture 01's open question — when is a score a probability.

The fraud example returns: a base rate of 1 in 1,000 and a handful of yes/no red flags, plus the house prices (flats of 40–120 m², and the 600 m² villa) of Lecture 01.

Argument Map



Speaker notes

Top row: classical regression, least squares as a likelihood, and what keeping the peak drops. Middle row: logistic regression as the Bernoulli twin, the comparison with naive Bayes, and what modelling only p(y | x) drops. Bottom: calibration and the takeaway that every loss so far is a likelihood. The strip marks Lecture 04's place in the arc. Return to the takeaway box at the end.

Classical Linear Regression

Guiding Question: You have fitted lines by least squares many times. What exactly is the method, and what justifies it?

Speaker notes

Shared units from the Data Science course, included unchanged. Skim what the room knows; slow down on the LMS update (error × feature, it returns for logistic regression) and on Gauss–Markov, whose "no distribution needed" is the contrast for Module 2.

Classification versus Regression

Classification

Y is a finite set of labels.

Will this customer default — yes or no?

Regression

$Y = \mathbb{R}$, the target is a quantity.

What credit line should this customer receive?

The learning machinery is the same; only the shape of Y — and therefore the loss — changes.

Speaker notes

Keep the same running example across both to make the point that the task, not the data, decides which you are doing.

The Model

Linear regression predicts the target as a weighted sum of the features:

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + \ldots + w_p x_p = \mathbf{w}^\top \mathbf{x}$$

with $x_0 = 1$ absorbing the intercept.

The hypothesis space is $\mathbb{R}^{p+1}$: one hypothesis per weight vector.

Speaker notes

Name it as a hypothesis space — here a continuous space of finite dimension, which is why calculus rather than enumeration finds the optimum.

The Objective: Residual Sum of Squares

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}\in\mathbb{R}^{p+1}} \mathrm{RSS}(\mathbf{w}), \qquad \mathrm{RSS}(\mathbf{w}) = \sum_{i=1}^{n} \left(y_i - \mathbf{w}^\top \mathbf{x}_i\right)^2$$

RSS measures **effectiveness** — how good the predictions are. Do not confuse it with **efficiency**, which is about runtime and computational effort.

Speaker notes

The effectiveness/efficiency distinction is from the source and worth keeping; students routinely say "efficient model" when they mean "accurate model".

One Feature: The Closed Form

For a single feature the minimiser can be written down directly:

Slope

$$\hat{w}_1 = \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sum_i (x_i - \overline{x})^2}$$

The covariance of x and y over the variance of x.

Intercept

$$\hat{w}_0 = \overline{y} - \hat{w}_1 \overline{x}$$

The fitted line passes through the point of means.

Speaker notes

Both formulas are built from quantities the students defined in the statistics module — covariance and variance. Say that explicitly: regression is not new mathematics, it is the old summary statistics arranged to answer a prediction question.

How Much Variance Is Explained?

$$\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}, \qquad R^2 = \frac{\mathrm{ESS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}$$

where TSS is the total, ESS the explained and RSS the residual sum of squares (the split holds for a least-squares fit with an intercept).

R^2 is the fraction of the target's variation the model accounts for: 0 means no better than predicting the mean, 1 means an exact fit.

Training R^2 never decreases when a feature is added, however useless the feature. A high training R^2 is not evidence that a model generalises — test R^2 and adjusted R^2 can and do fall.

Speaker notes

$R^2 = 0$ is exactly the trivial baseline that always predicts the mean. The warning is the regression counterpart of training error — and the reason a model is judged on data it was not fitted to.

Higher Dimensions

With p features the same objective in matrix form,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the optimum in one step whenever $\mathbf{X}^\top \mathbf{X}$ is invertible.

> Invertibility fails when features are linearly dependent or when $p > n$ — more unknowns than equations. Both are common in practice; regularisation is the standard remedy.

Speaker notes

Do not derive the normal equations; state them and spend the time on when they break. The p > n case is the modern one — genomics, text — and motivates regularisation more than the algebra does.

Takeaway

Linear regression fits a weighted sum by minimising squared residuals. Its solution is built from **covariance over variance**, and R^2 reports how much variation it explains — not whether it will generalise.

Linear in the Parameters, Not in the Features

A model is **linear** when it is linear in the **weights** $\mathbf{w}$. The features may be arbitrary transformations of the input.

Plain

$$h(\mathbf{x}) = w_0 + w_1 x$$

a straight line

Polynomial

$$h(\mathbf{x}) = w_0 + w_1 x + w_2 x^2$$

a curve — still a *linear model*

Basis functions

$$h(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x})$$

any fixed feature map ϕ

What *linear* forbids: a transformation that has to be learned from the data. The feature map is fixed before fitting; only the weights are learned.

Speaker notes

This is the slide that unlocks the rest: the same least-squares machinery fits curves, interactions and periodic patterns, as long as the transformation is chosen in advance. Ask what is *not* expressible before revealing the note — anything where the transformation itself must be learned, which is what neural networks do.

Consequences

The good

The closed-form solution and all statistical guarantees carry over unchanged.

The catch

The feature map is a **modelling decision**. Choose too rich a basis and the model overfits; too poor and it underfits.

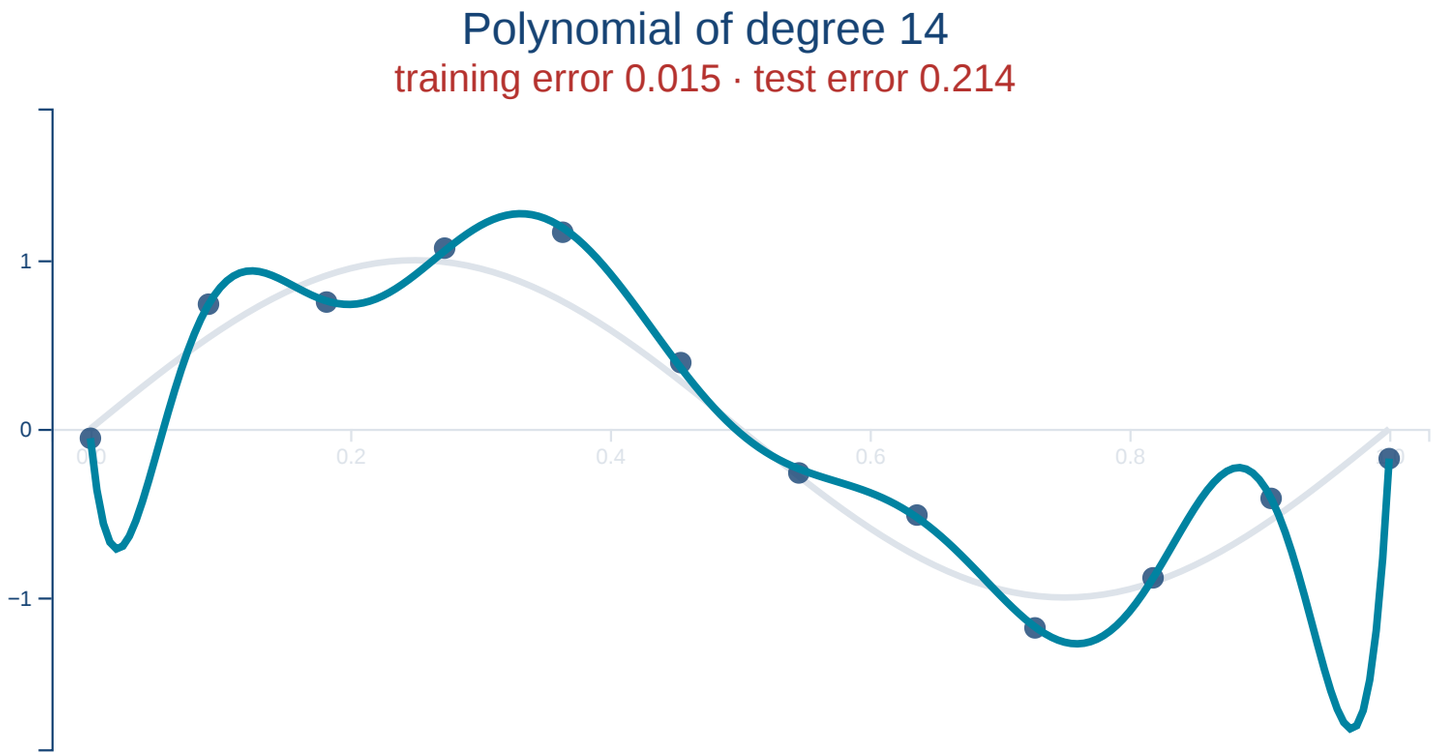
> Increasing the polynomial degree turns a dial on the bias–variance axis.

Speaker notes

The basis is the inductive bias made explicit. This is the cleanest example in the course of choosing a hypothesis space deliberately.

Turning the Capacity Dial

Linear in the parameters, any basis in the features: a polynomial of degree k is still "linear regression" — and k is a dial.



Speaker notes

Basis functions: linear in the parameters, any shape in the features. Each click raises the degree: training error falls monotonically, test error bottoms out and climbs again. In today's language: every degree is fitted by keeping the peak; the penalty of Module 5 (a prior) is the other way to turn the dial down.

Fitting h by Repeated Correction

LMS (least mean squares)

1. Initialise the weights w.
2. Pick a training example $(\mathbf{x}, y)$ from the training set D.
3. Compute the prediction $h_w(\mathbf{x})$ and the error $e = y - h_w(\mathbf{x})$.
4. Update every weight: $w_i \leftarrow w_i + \eta e x_i$.
5. Repeat 2–4 until the error on D stops decreasing.

Speaker notes

Read the update aloud in words: move each weight in the direction that would have reduced this example's error, in proportion to how much that feature contributed. That sentence is all of gradient descent, before any calculus.

What the Update Rule Encodes

Direction

The sign of e decides whether the prediction goes up or down.

Attribution

The factor x_i gives each feature credit proportional to its value.

Step size

The learning rate η trades speed against stability.

Too large an η oscillates or diverges; too small an η converges slowly. It is a hyperparameter — chosen, not learned.

Speaker notes

Ask what happens with η far too large before showing it; the prediction that "it overshoots and gets worse" is usually made correctly, which earns the hyperparameter concept. The visualisation in the optimisation unit shows it.

Two Ways to the Same Minimum

Direct (closed form)

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Exact, one step. Requires forming and inverting a $(p+1) \times (p+1)$ matrix — costly for large p, impossible when it is singular.

Iterative (gradient descent)

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \mathrm{RSS}(\mathbf{w})$$

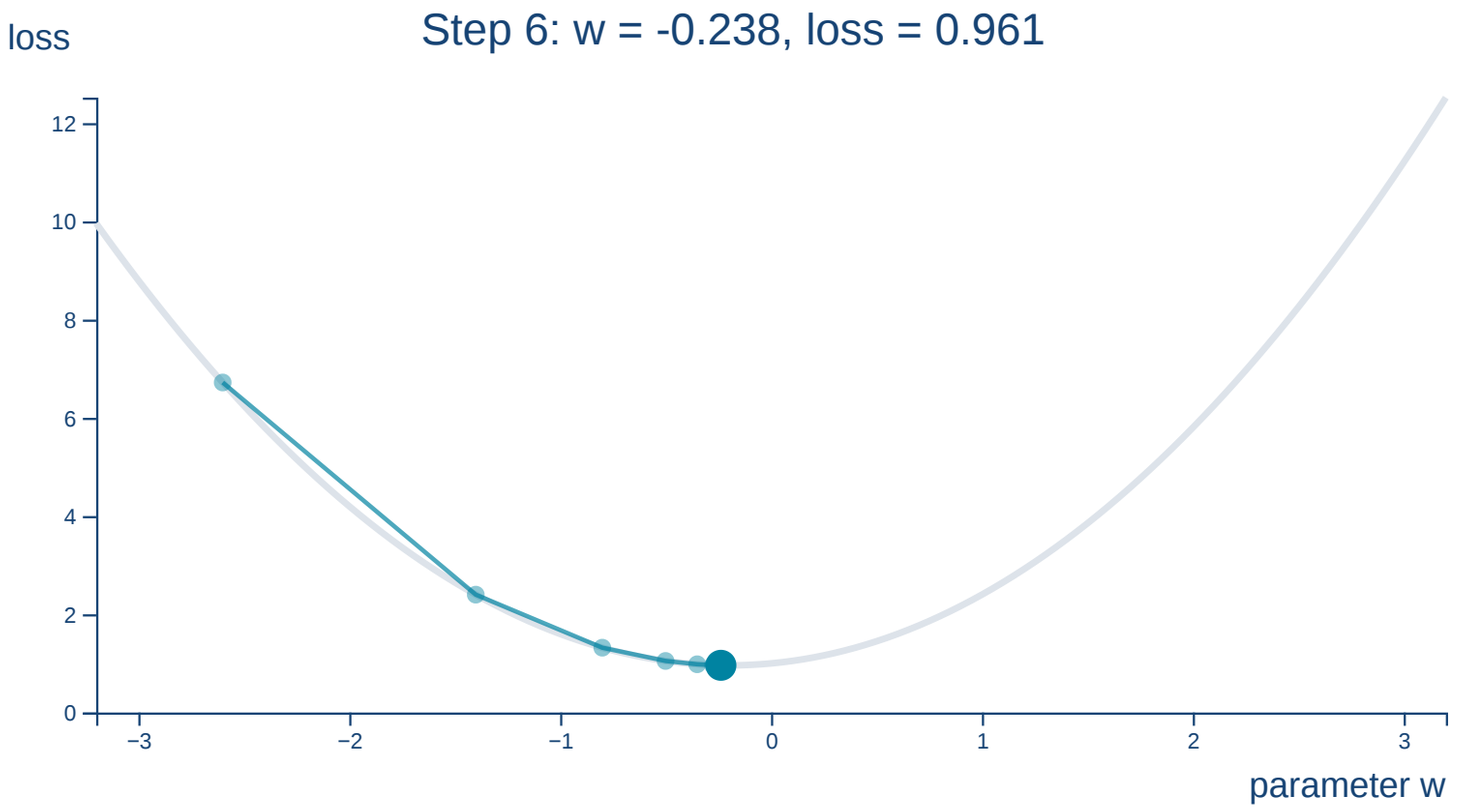
Approximate, many steps. Scales to large p and n, works when the direct solution does not exist, and extends to models with no closed form at all.

The iterative rule is exactly the **LMS algorithm**: its update $w_i \leftarrow w_i + \eta e x_i$ *is* a gradient step on RSS, taken per example instead of per dataset (the factor 2 is absorbed into η).

Speaker notes

The right-hand column is the one that generalises: logistic regression and neural networks have no closed form, so the iterative method is the one that generalises. Make the LMS identification explicit — students otherwise carry two unconnected procedures.

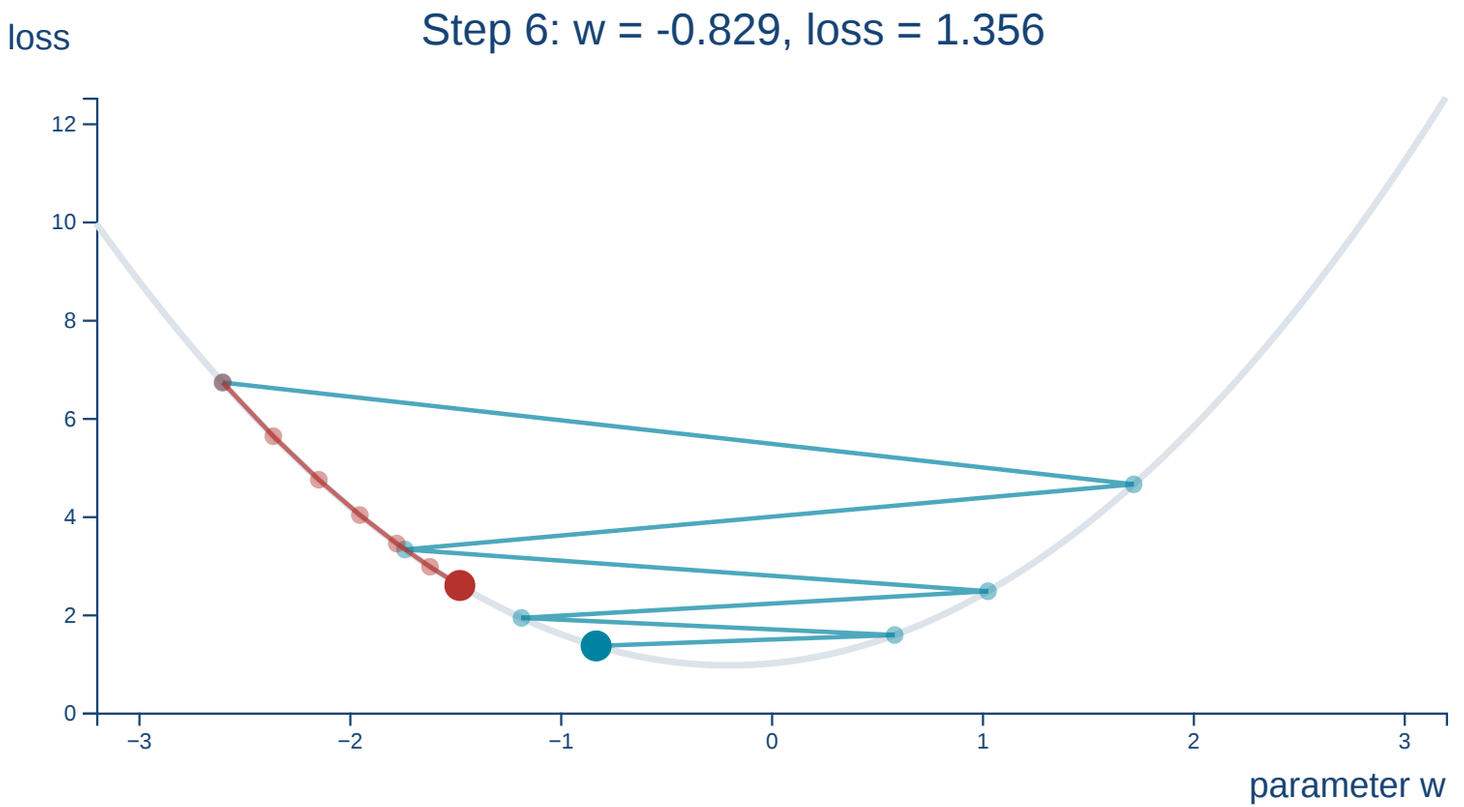
Watching Gradient Descent Work



Speaker notes

Stepped viz (P6). Each click is one update. Ask before the first click where the steps will be largest — where the slope is steepest, far from the minimum — and why they shrink as the minimum approaches, without η changing.

The Learning Rate Decides Everything



Teal: $\eta = 0.9$ — overshoots and oscillates across the minimum. Red: $\eta = 0.05$ — stable, but slow: after six steps it has covered about half the distance.

The learning rate is a **hyperparameter**: it is chosen, not learned, and both failure modes are silent — one looks unstable, the other looks converged.

Speaker notes

Predict-then-check: ask which curve belongs to which η before revealing. The "looks converged" failure is the dangerous one — a slow run and a finished run produce the same-looking loss curve over a short window.

Takeaway

> Use the **closed form** when the matrix is small and well-conditioned; use **gradient descent** when it is not — and for most models beyond linear regression, no closed form exists at all.

The Question Nobody Asks

We minimised the *squared* residuals. Why squares — why not absolute deviations, or fourth powers?

Speaker notes

Guiding question. Most students have never been given a reason beyond differentiability. There is a real one, and it is worth the slide.

Geometric View

The predictions $\mathbf{Xw}$ live in the **column space** of $\mathbf{X}$ — everything the features can express. The observed $\mathbf{y}$ generally does not.

Least squares picks the point of that space **closest to $\mathbf{y}$ in Euclidean distance**: the orthogonal projection. The residual is perpendicular to every feature.

Consequence

The residual carries no remaining linear information about the features.

Speaker notes

The orthogonality condition is the geometric content of the normal equations. It also explains why adding a feature can never increase RSS: the space you project onto only grew.

The Gauss–Markov Theorem

Gauss–Markov (informal). Among all estimators that are

- **linear** in the observed outputs $(y_1, \ldots, y_n)$, and
- **unbiased**,

the least-squares estimator has the **lowest variance**. It is the *Best Linear Unbiased Estimator* — BLUE.

Best

lowest variance among the candidates

Linear

a linear function of the observations

Unbiased

right on average over repeated samples

Speaker notes

Spell out what the theorem does *not* say: nothing about non-linear or biased estimators — and biased estimators (ridge, lasso) can beat OLS in prediction error. That is not a contradiction, it is a different criterion.

What This Buys, and What It Costs

The payoff

Minimising RSS yields optimal statistical properties under **mild assumptions** — no normality needed, only a correctly specified linear model, linearly independent features, and zero-mean, uncorrelated errors of constant variance.

The price

Alternatives — absolute error, Huber loss — are **more robust to outliers** but give up the BLUE guarantee.

Squared error is not an arbitrary convenience: it is the loss whose minimiser is the best linear unbiased estimator. Departing from it is legitimate — but it is a trade, and you should be able to name what you traded.

Speaker notes

Close the loop with the robustness discussion from descriptive statistics: mean versus median reappears here as squared versus absolute loss. Same trade-off, new context — exactly the spiral the pedagogic concept asks for.

Least Squares Is a Likelihood

Guiding Question: You have minimised squared error many times. What did that choice assume about the world — and what does it leave out?

The Same Model, Told Differently

Classical account

Choose the line that **minimises squared error**:

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

Probabilistic account

State how a price **arises** for a given flat:

$$y = \mathbf{w}^\top \mathbf{x} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

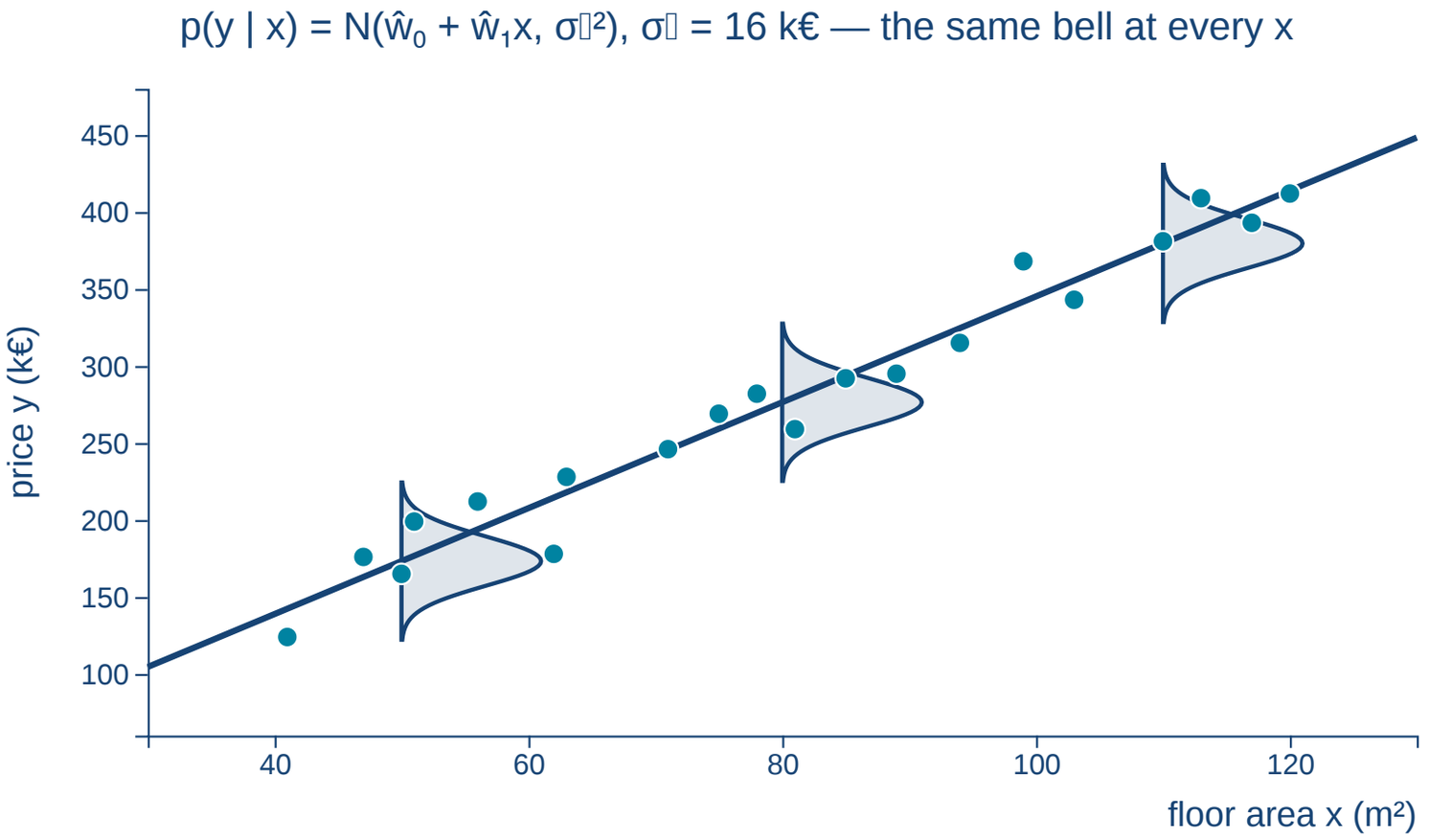
then keep the peak of the likelihood.

Do the two accounts give the same $\hat{\mathbf{w}}$ — and if so, what did the classical account assume without saying it?

Speaker notes

Module 1 has just given the classical account: squared error, closed form, Gauss–Markov. Question first (P1). Let students vote. Most will guess "same" without being able to say why; the next two slides give the picture and then the derivation. The second half of the question is the module's guiding question in small.

A Distribution at Every x



A **conditional distribution** $p(y \mid x) = \mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$: one bell per x, **equally wide**. Nothing about how x arises.

Speaker notes

Click 1: the fitted line, ŷ ≈ 1 + 3.44·x (k€, m²). Click 2: the bells at 50, 80 and 110 m², each with σ ≈ 16 k€. Read a bell aloud: "an 80 m² flat costs about 276 k€, give or take 30". This is the picture that makes both later points of the lecture visible: the loss is the shape of the bell (next slide), and the model is only about y *given* x — there is no bell for x itself. That second point is shortcut 2, named in Module 3: the model never learns what a typical flat looks like, so it cannot tell that a villa is unusual.

The Derivation

With $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)$ and i.i.d. rows:

$$\ell(\mathbf{w}) = \sum_{i=1}^{n} \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2}\right) = \underbrace{-\frac{n}{2}\log(2\pi\sigma^2)}_{\text{no } \mathbf{w}} - \frac{1}{2\sigma^2}\mathrm{RSS}(\mathbf{w})$$

> Maximising the Gaussian log-likelihood **is** minimising RSS: $\hat{\mathbf{w}}_{\mathrm{ML}} = \hat{\mathbf{w}}_{\mathrm{OLS}}$. Squared error is the exponent of the bell.

Speaker notes

Walk the line slowly — the course's thesis becomes concrete here. The first term does not contain w, the factor $1/(2\sigma^2)$ is positive, so the arg max over w is the arg min of RSS whatever σ is. Answer to the previous slide's vote: the same $\hat{w}$ — and the classical account had assumed, without saying so, Gaussian noise of one fixed width.

What Squared Error Assumed

the noise model says	squared error silently assumes	check on the residuals
ε has mean 0	the line is the right shape	no pattern against x
one σ for every x	**homoscedastic** noise	no fan shape
rows independent	i.i.d. (Lecture 02)	no pattern in time or space
ε Gaussian	thin tails: large errors are rare	no heavy tails, few outliers

Choosing a loss and choosing a noise model are **one decision**. Squared error ⇔ Gaussian noise; absolute error ⇔ Laplace noise — heavier tails, so a large residual is less surprising and an outlier pulls the fit less.

Speaker notes

Each row turns a property of the loss into a claim about the world, and every claim can be checked with a residual plot — that is what stating the model buys. The Laplace row answers a question students may bring from earlier courses ("why is L1 loss robust?") with a reason instead of a rule: −log of a Laplace density is |ε|/b plus a constant.

The Peak of σ^2

The same likelihood also has a peak in σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\mathrm{RSS}}{2\sigma^4} = 0 \quad \Longrightarrow \quad \hat{\sigma}^2_{\mathrm{ML}} = \frac{\mathrm{RSS}}{n}$$

Which of Lecture 02's two variances is this — and is it too small or too large?

Speaker notes

A second, quiet cost of keeping the peak (Lecture 03). Lecture 02 divided by n − 1 because one mean had been estimated; here p weights have been, so E[RSS] = (n − p)σ² and E[σ̂²_ML] = (n − p)/n · σ². With 20 flats that is 10% in the variance; with 5 flats and 2 weights it would be 40%. Small or not, it always points the same way: the peak makes the model look less noisy than it is.

The Peak of σ^2

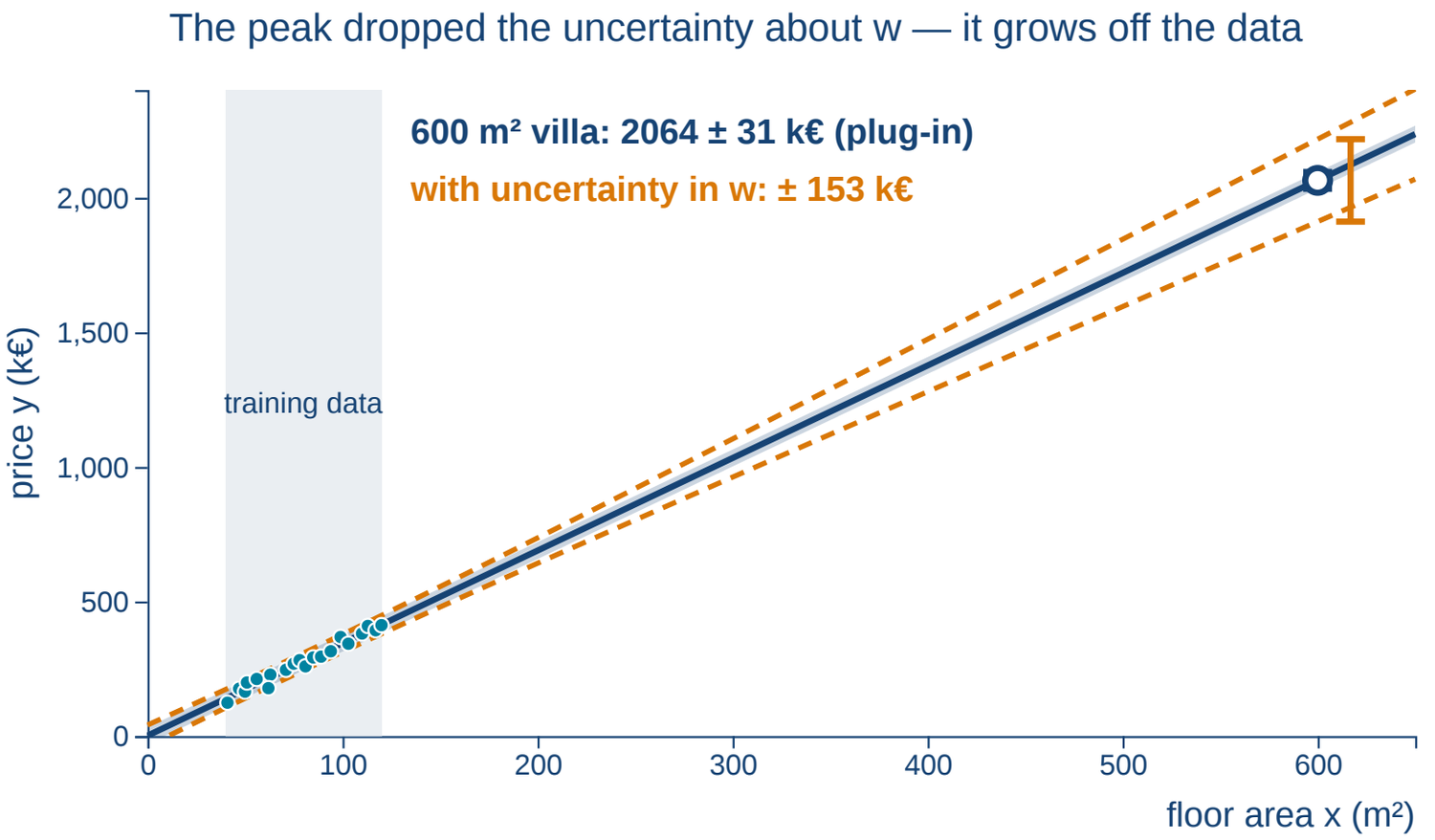
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Which of Lecture 02's two variances is this — and is it too small or too large?

> The $1/n$ one. The line was fitted to the same data, so residuals are smaller than the true errors: $\hat{\sigma}^2_{\mathrm{ML}}$ is **biased low**. The unbiased version divides by $n - p$ for p fitted weights. Houses: $n = 20$, $p = 2$, RSS $\approx 5{,}097$ $\rightarrow$ $\hat{\sigma} \approx 16.0$ instead of 16.8 k€.

The Villa Again



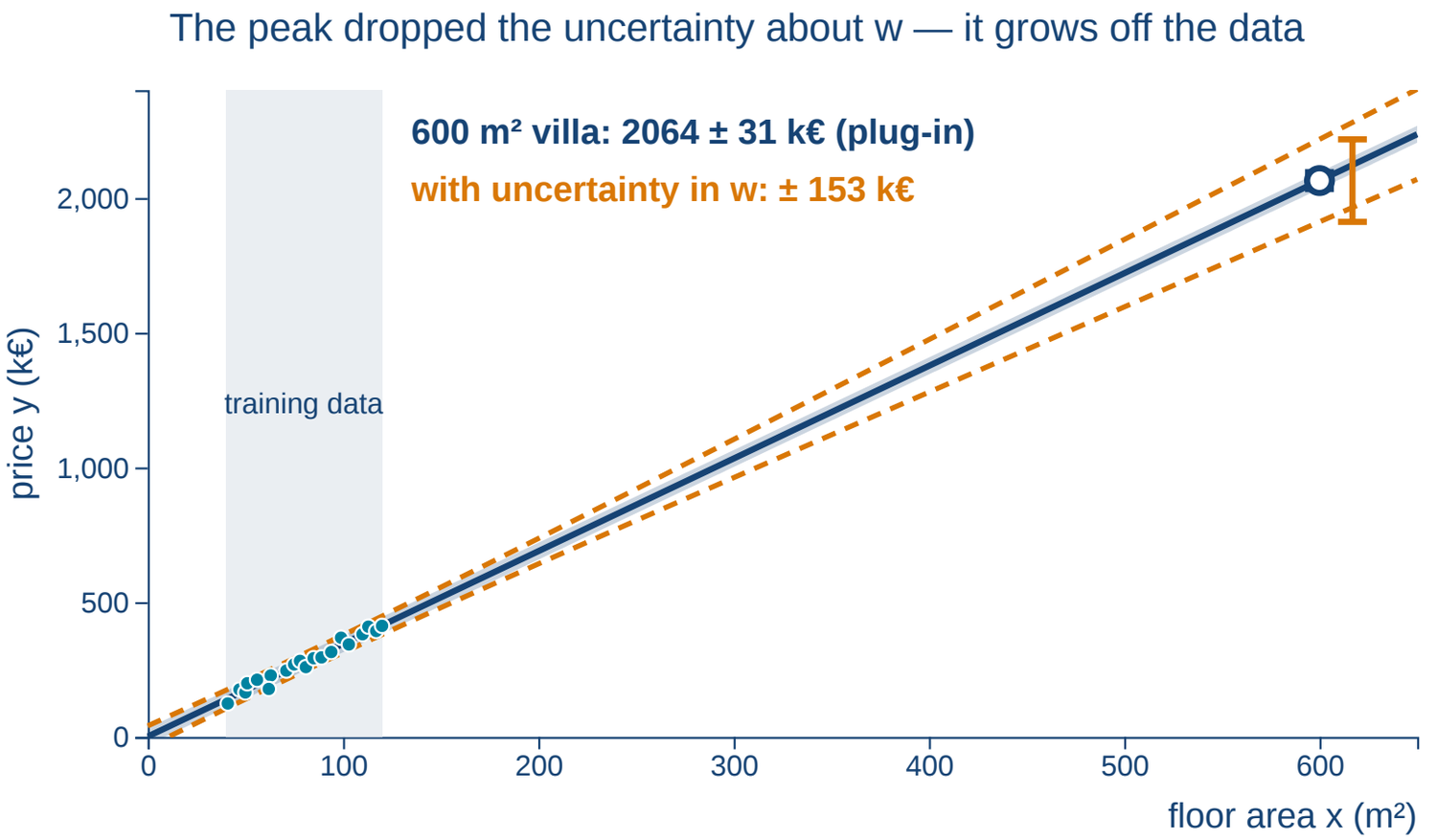
The band is as narrow at the villa as at a flat. Is the model as sure?

Speaker notes

Step 0: the fit and the plug-in band $\hat{y} \pm 1.96 \cdot \sigma$̂ = ± 31 k€. Click 1: the villa of Lecture 01 — predicted about 2,064 k€ ± 31 k€, the same half-width as for any flat. Ask the question before the second click. Click 2: the band that also counts the uncertainty about the two weights, $1.96 \cdot \sigma$̂$\cdot\sqrt{(1 + 1/n + (x - x$̄$)^2/\Sigma(x_i - \bar{x})^2)}$: ± 32 k€ at 80 m², ± 153 k€ at 600 m². This is Lecture 01's epistemic uncertainty, made visible — and exactly the spread that Lecture 03 said the peak drops.

One more honesty point, for the discussion: even the wide band trusts the straight line all the way to 600 m². Whether villas follow the price per m² of flats is a modelling question no band can answer.

The Villa Again



The band is as narrow at the villa as at a flat. Is the model as sure?

No: ± 31 k€ is only the noise $\hat{\sigma}$. The uncertainty about $\mathbf{w}$ — dropped with the peak — grows off the data: ± 153 k€ at the villa (Lecture 07).

Mini-Exercise

The residuals of your house-price model fan out: about ± 10 k€ for 40 m² flats, ± 60 k€ for 120 m² flats.

Which assumption fails — and which output of the model can you no longer trust?

Speaker notes

One minute to read, one to think. Students with a statistics background may add that $\hat{w}$ stays unbiased but is no longer the most efficient estimator (the Gauss–Markov condition "equal variance" fails) — correct, and worth one sentence. The fix, if asked: model σ as a function of x, which is again just stating a different noise model.

Mini-Exercise

The residuals of your house-price model fan out: about ± 10 k€ for 40 m² flats, ± 60 k€ for 120 m² flats.

Which assumption fails — and which output of the model can you no longer trust?

One σ for every x (homoscedasticity). The line may still be fine, but the **band** is not: a single $\hat{\sigma}$ is too wide for small flats and too narrow for large ones.

Takeaway

Least squares **is** maximum likelihood under Gaussian noise: the loss states the noise model, and the model is a distribution $p(y \mid x)$. It is also a **peak**: $\hat{\sigma}^2$ divides by n, and the band is as narrow at the villa as at a flat — the uncertainty about $\mathbf{w}$ was dropped.

Classification: Model Only p(y | x)

Guiding Question: Lecture 01's fraud model said 0.55. Where does such a number come from — and what does the model refuse to know?

Why Not Regress on 0/1?

Code fraud as $y = 1$, legitimate as $y = 0$, and fit Module 2's line.

Speaker notes

A known failure, stated in the language of Module 2: the loss is wrong because the noise model is wrong. A Gaussian around $w^T x$ is a model of a real number; fraud is a yes/no outcome. Everything else in the module follows from replacing the Gaussian by the Bernoulli.

Why Not Regress on 0/1?

Code fraud as $y = 1$, legitimate as $y = 0$, and fit Module 2's line.

Squared error punishes being right

A transaction far on the correct side has a large residual, so least squares drags the boundary towards cases it already classifies correctly.

The output is not a probability

$\mathbf{w}^\top \mathbf{x}$ can be 17 or -40. The Gaussian bell of Module 2 puts mass on every real y — for a label that is 0 or 1.

Keep the linear score, change what it models: the **log-odds**, and a noise model that produces 0s and 1s.

Log-Odds Are Linear

$$\ln \frac{P(y=1 \mid \mathbf{x})}{1-P(y=1 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x} \qquad \Longleftrightarrow \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1+e^{-\mathbf{w}^\top \mathbf{x}}}$$

read the weights as	fraud example
the intercept: the log-odds when every feature is 0	no features at all: $w_0 = \ln \frac{1}{999} \approx -6.9$, the base rate 1 in 1,000
a weight: what one unit of a feature **adds** to the log-odds	"new device" with $w = \ln 6 \approx 1.8$: odds $\times 6$
the boundary: where $\mathbf{w}^\top \mathbf{x} = 0$	score 0.5 — still a straight line

Speaker notes

The sigmoid is not an arbitrary squashing function; it is the inverse of the logit. The reading of the intercept ties the model to Lecture 02: a fraud model with no features has its peak at the base rate (h/n again: 1 in 1,000, log-odds ln(1/999)). The weight reading is the one to stabilise: weights add log-odds, so they multiply odds — the same bookkeeping as likelihood ratios in Bayes' theorem, which naive Bayes (Lecture 03) already exploited.

Where Does Cross-Entropy Come From?

$$L(\mathbf{w}) = -\sum_{i=1}^{n} \Big[y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \Big], \qquad h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$$

You have minimised this loss many times. Which noise model does it state?

Speaker notes

Math first (P5): the loss on the screen, the question, then the reveal. It is the same derivation as h/n in Lecture 03, with θ replaced by h(x) for every row. Stress the last sentence: this is a *conditional* story, y given x. It is not a generative story — nothing in it could produce a transaction.

Where Does Cross-Entropy Come From?

$$L(\mathbf{w}) = -\sum_{i=1}^{n} \Big[y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \Big], \qquad h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$$

You have minimised this loss many times. Which noise model does it state?

Given $\mathbf{x}$, the label is a **Bernoulli** draw with bias h: $P(y_i \mid \mathbf{x}_i, \mathbf{w}) = h_i^{y_i}(1 - h_i)^{1 - y_i}$. With i.i.d. rows,

$$\ell(\mathbf{w}) = \sum_{i=1}^{n} \Big[y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \Big] = -L(\mathbf{w})$$

> Cross-entropy **is** the Bernoulli negative log-likelihood — Lecture 03's coin, biased by $\mathbf{x}$. How $\mathbf{x}$ arises is never said.

The Same Update Again

With $\sigma' = \sigma(1-\sigma)$, the gradient of the log-likelihood is

$$\frac{\partial \ell}{\partial \mathbf{w}} = \sum_{i=1}^{n} (y_i - h_i)\mathbf{x}_i$$

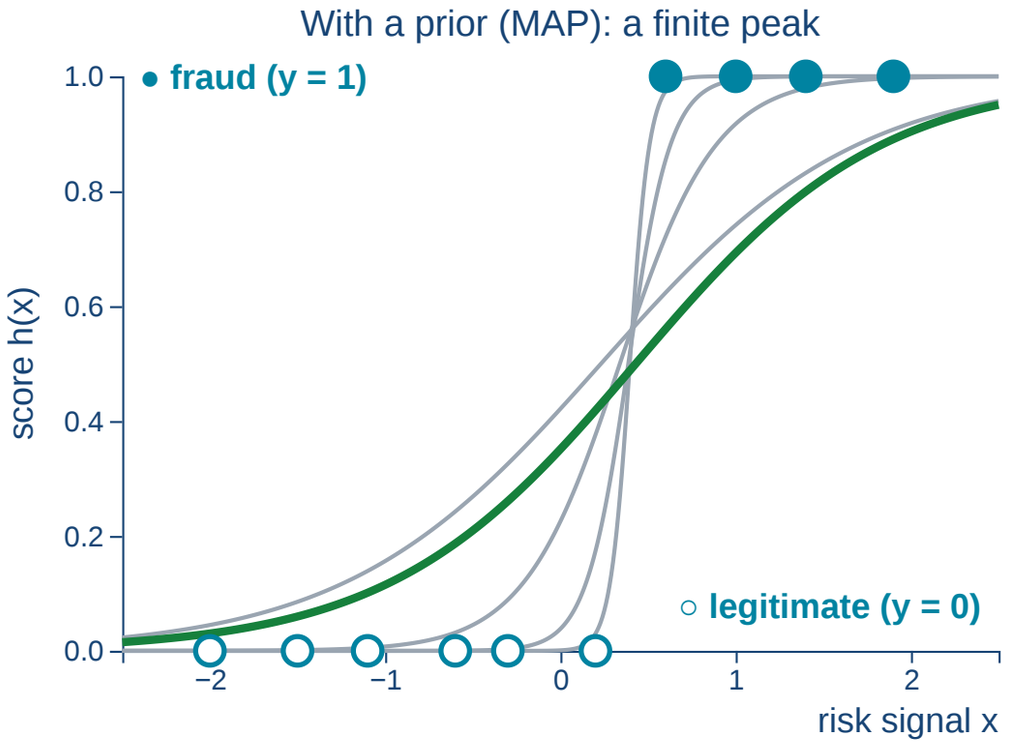
model	noise	loss	gradient of ℓ
linear regression	Gaussian	squared error	$\frac{1}{\sigma^2}\sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)\mathbf{x}_i$
logistic regression	Bernoulli	cross-entropy	$\sum_i (y_i - \sigma(\mathbf{w}^\top \mathbf{x}_i))\mathbf{x}_i$

Error × feature — the LMS update with a sigmoid in the prediction; no closed form, so iterate. A fraud scored 0.01 costs $-\ln 0.01 \approx 4.6$, scored 0.99 only 0.01: confident mistakes dominate.

Speaker notes

One line of calculus: d/dw [y ln h + (1 − y) ln(1 − h)] = (y/h − (1 − y)/(1 − h)) · h(1 − h) · x = (y − h) x. The table is the unification the lecture is after: two noise models, one form of update. The per-example version is exactly LMS with ŷ replaced by σ($w^T x$).

When the Peak Runs Away



steps	slope	log-lik
10	1.4	-2.68
100	3.6	-1.04
1,000	8.3	-0.36
10,000	17.8	-0.06
MAP	**1.4**	**-2.54**

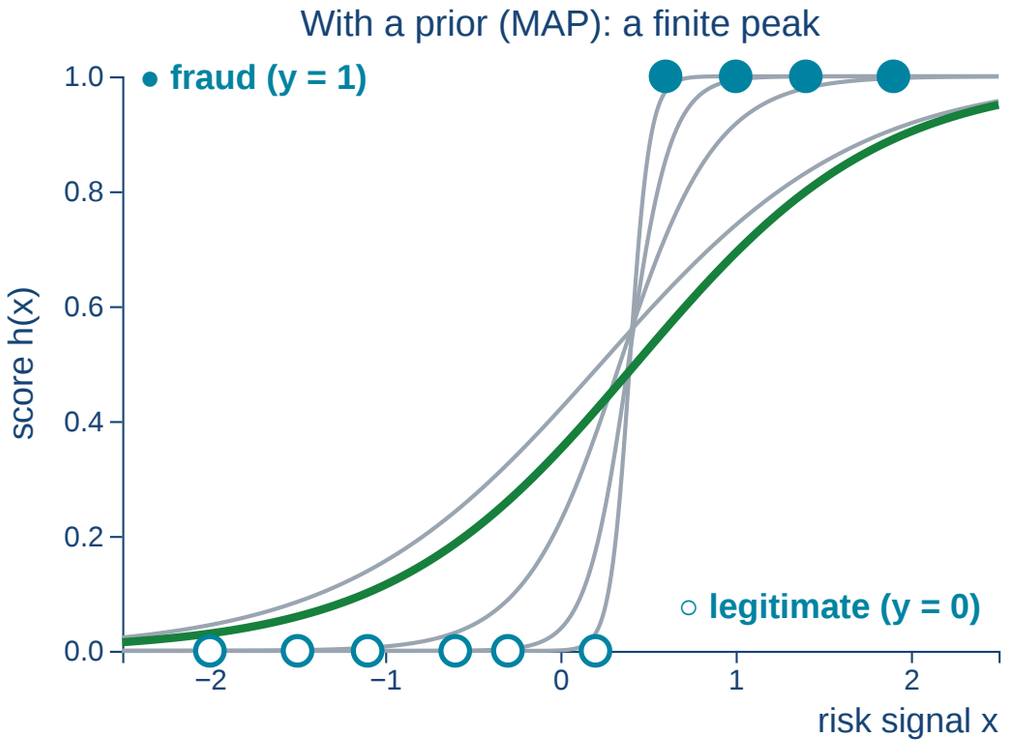
prior on the slope: N(0, 1)
stays, however long it runs

Every fraud scores above every legitimate case. Where is the peak of the likelihood?

Speaker notes

Ask the question before the first click and let students guess. Clicks 1–4: the curve after 10, 100, 1,000 and 10,000 gradient steps — slope 1.4, 3.6, 8.3, 17.8, log-likelihood −2.68, −1.04, −0.36, −0.06. It never stops; the optimiser is not broken, the peak does not exist. Click 5: a Gaussian prior N(0, 1) on the slope; MAP converges to slope 1.4 and stays there. In practice every library adds this prior by default — it is called regularisation, and Module 5 reads it as a prior. Separable data is common with many features (rare fraud, thousands of attributes), so this is not an exotic case.

When the Peak Runs Away



steps	slope	log-lik
10	1.4	-2.68
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10,000	17.8	-0.06
MAP	**1.4**	**-2.54**

prior on the slope: N(0, 1)
stays, however long it runs

Every fraud scores above every legitimate case. Where is the peak of the likelihood?

Nowhere. The likelihood keeps rising as the slope grows; the scores harden to 0 and 1 — three heads in three flips. A prior's peak (MAP) is finite.

Shortcut 2: Model Only $p(y \mid \mathbf{x})$

Linear and logistic regression are **discriminative**: a model of y given $\mathbf{x}$, none of $\mathbf{x}$.

What is dropped: $p(\mathbf{x})$

- a transaction unlike any seen still gets a confident score
- a missing field cannot be averaged out — it must be filled in
- no way to simulate a plausible transaction

What is gained

- no assumptions about how features are distributed — where, with hundreds of features, most modelling errors hide
- every parameter serves the decision

Lecture 03 took the other route: model the joint $p(\mathbf{x}, y)$ and condition — **generative**. Module 4 compares the two.

Speaker notes

Name the shortcut explicitly and tie it to the L01 roadmap row "only what you need". The first dropped item is the villa of Module 2 in classification form: the model cannot know that an input is unusual, because it never modelled inputs. The gained column is not a consolation prize: for prediction with many features, discriminative models are usually more accurate — Module 4 makes the comparison with naive Bayes.

Mini-Exercise

Your fraud model scores **0.97** for a transfer 50 times larger than any in its training data.

Should you trust the 0.97 — and could the model itself warn you?

Speaker notes

One minute to read, one to think. The transfer is the classification twin of the 600 m² villa. Good follow-up if time allows: "which of Lecture 01's two kinds of uncertainty is this?" — epistemic; ten such transfers in the training data would help.

Mini-Exercise

Your fraud model scores **0.97** for a transfer 50 times larger than any in its training data.

Should you trust the 0.97 — and could the model itself warn you?

Not blindly: the linear log-odds simply **extrapolate**, like the line at the villa. And no — the model has **no** $p(\mathbf{x})$, so nothing in it can see that the amount is unusual. That needs a model of $\mathbf{x}$ or a separate check.

Takeaway

Logistic regression is Module 2 with Bernoulli noise: linear log-odds, cross-entropy as the negative log-likelihood, error × feature as the gradient. It **keeps the peak** — which runs away on separable data — and **models only** $p(y \mid \mathbf{x})$: it never learns what a normal transaction looks like.

Generative vs Discriminative

Guiding Question: Naive Bayes and logistic regression see the same fraud table. Which would you trust with 50 labelled frauds — and which with 50,000?

The Same Line, Fitted Twice

Take the logarithm of the naive-Bayes odds. For yes/no flags $x_j \in \{0, 1\}$ with $\theta_{jy} = P(x_j = 1 \mid y)$:

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = w_0 + \sum_{j=1}^{p} w_j x_j, \qquad w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Where have you seen a log-odds that is linear in x?

Speaker notes

Math first (P5). The derivation is two lines and belongs on the exercise sheet: log p(x_j | y) = x_j log θ_jy + (1 − x_j) log(1 − θ_jy), so each flag contributes a constant plus x_j times a weight; the constants and the log prior odds collect in w_0. With the fraud numbers, w for the new device is log 6 − log(0.4/0.9) ≈ 1.79 + 0.81 = 2.60.

The consequence of "jointly": logistic regression sees that the new device and the foreign IP arrive together and splits the credit between them. Naive Bayes cannot; each weight is estimated as if its flag were alone. That is the double counting of Lecture 03, now visible in the weights.

The Same Line, Fitted Twice

Take the logarithm of the naive-Bayes odds. For yes/no flags $x_j \in \{0, 1\}$ with $\theta_{jy} = P(x_j = 1 \mid y)$:

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = w_0 + \sum_{j=1}^{p} w_j x_j, \qquad w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Where have you seen a log-odds that is linear in x?

> **Logistic regression** (Module 3): $P(\text{fraud} \mid x) = \sigma(w_0 + w^\top x)$. Same family of classifiers — naive Bayes sets the weights by **counting**, one attribute at a time; logistic regression by **maximising** $\sum_i \log p(y_i \mid x_i)$, all weights jointly.

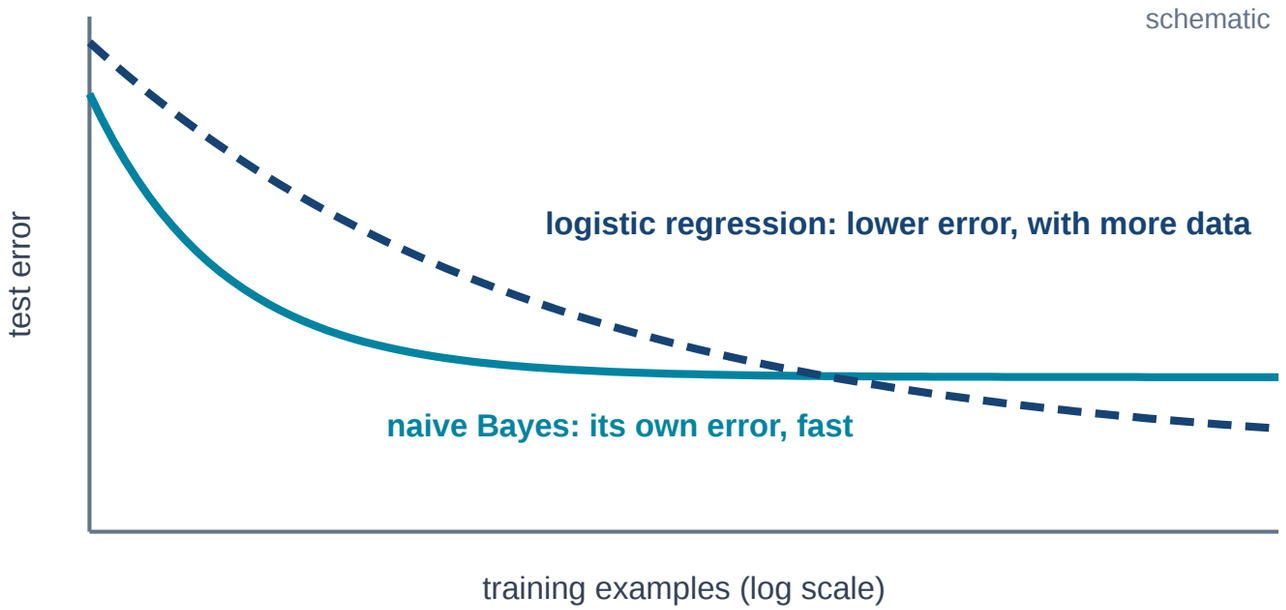
Two Ways to a Classifier

	naive Bayes — generative	logistic regression — discriminative
models	$P(y)$ and $p(x \mid y)$	$P(y \mid x)$ only
fitted by	counting (joint likelihood)	iterative optimisation (conditional likelihood)
correlated flags	counted twice	credit shared
a missing field	drop its factor	needs a value
an unusual x	low $p(x)$: can be flagged	still a confident score
a new base rate	change $P(y)$	refit or correct the intercept
drops	dependences between attributes	everything about x

Speaker notes

A table because the two are compared along shared dimensions. Walk it row by row and let students predict each cell for logistic regression before showing it, if time allows. The last row is the course arc: each model is placed by what it drops. "Correct the intercept": in logistic regression the base rate sits inside w_0, so a new base rate shifts w_0 by the log of the ratio of prior odds — possible, but not visible in the model as a separate factor.

Few Data, Many Data



Few labelled frauds: naive Bayes. Its assumption acts like a strong prior — little to estimate, so it settles fast.
Many labelled frauds: logistic regression. It pays for fewer assumptions with data, and ends lower.

Shown empirically and in theory by Ng & Jordan (2002). A wrong assumption costs accuracy in the limit and buys it early — the prior of Lecture 02, again.

Speaker notes

The figure is schematic — say so. Ng & Jordan (NIPS 2002, "On discriminative vs. generative classifiers") show that naive Bayes approaches its asymptotic error with a number of examples logarithmic in the number of attributes, logistic regression with a number linear in it; the discriminative asymptote is never worse. On many real datasets the curves cross.

Answer the module's guiding question: with 50 labelled frauds, trust naive Bayes' ranking; with 50,000, logistic regression. And the general point: an assumption is information. When it is roughly right it substitutes for data; when it is wrong it caps what data can do.

Mini-Exercise

Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

Speaker notes

The generative advantage in its most practical form. Marginalising a missing attribute out of a product of per-attribute factors is summing one factor over its values, which gives 1. The number: 15/1014 ≈ 0.0148. Compare with 17% when the IP was present and foreign: the missing field matters, and naive Bayes reports that honestly instead of guessing it.

Mini-Exercise

Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

Naive Bayes drops the factor: $\sum_v p(x_{\text{IP}} = v \mid y) = 1$, so the odds become $6 \cdot 2.5 : 999$ — about **1.5%**. **Logistic regression** has no model of x: it needs *some* value for x_{IP}, so someone must impute one, or train a second model without the field.

Takeaway

> Naive Bayes and logistic regression draw **the same kind of line**; they differ in what they model. The generative model pays with assumptions about x and gets answers about x; the discriminative one pays with data.

lecture	shortcut	what is dropped
03	keep the peak	how sure we are
03	assume it apart	every dependence between attributes
04	only $p(y \mid x)$	every question about x

Naive Bayes is a graph with one hub. **Lecture 05** draws the others — and keeps exactly the dependences we claim.

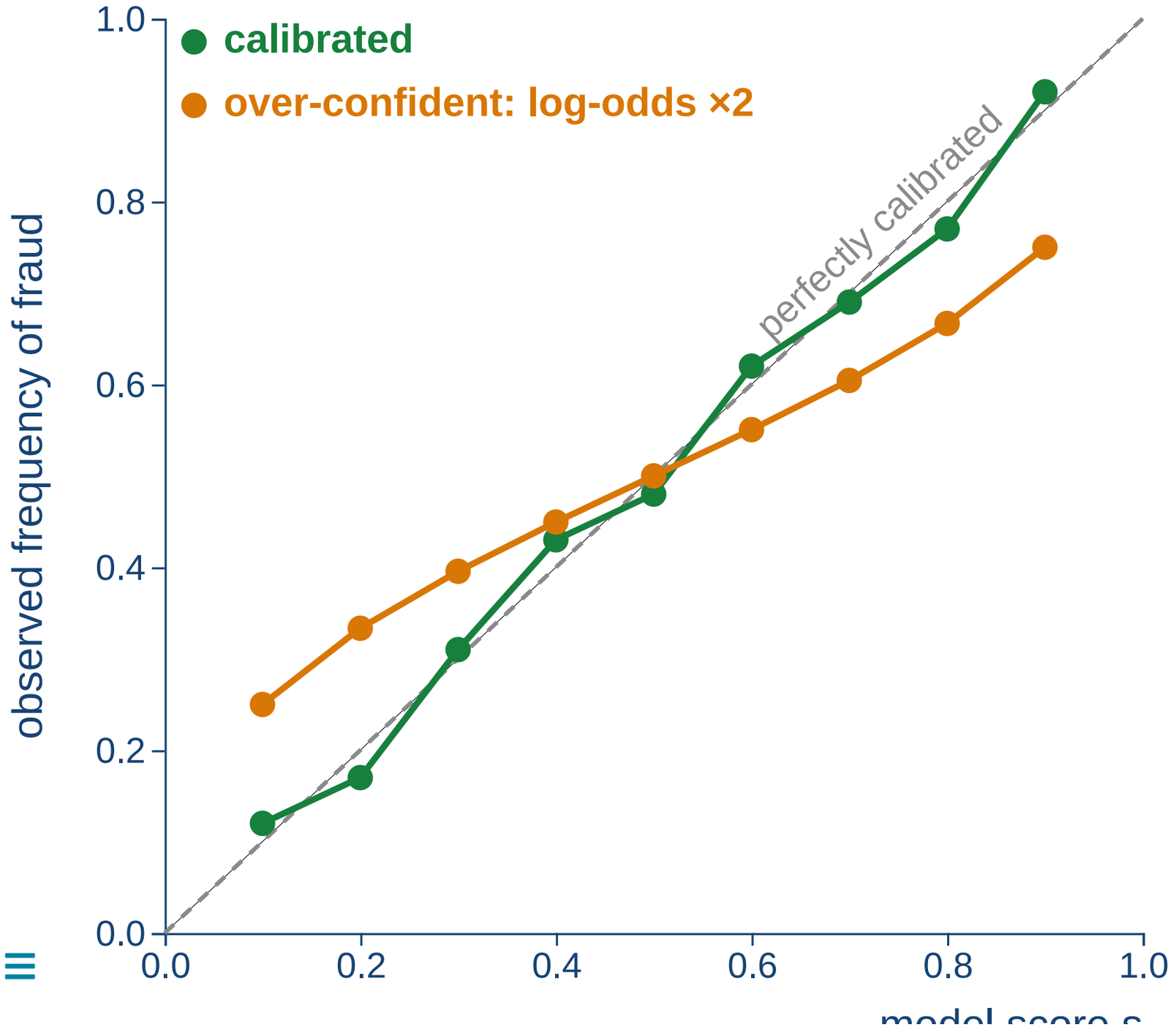
Speaker notes

Close the lecture on the arc. The table is the L01 shortcut table, now with three rows filled. The teaser: the star graph of Lecture 03 is the simplest graphical model; Lecture 05 lets the modeller choose which arrows to keep, so the assumption becomes a statement rather than a blanket.

When Is a Score a Probability?

Guiding Question: Lecture 01 promised to say when a model's output earns the name "probability". When does it — and what breaks it?

When Is 0.7 a Probability?



A score is **calibrated** if, among all cases scored s, a fraction s is positive. A **reliability diagram** plots that fraction against s.

Predict before click 2: a model whose log-odds are twice too large — above or

Speaker notes

below the diagonal at high scores?

Step 0: the diagonal. Click 1: a calibrated model — the points scatter around the diagonal, as bins of real data would. Ask for the prediction, then click 2: the over-confident model lies *below* the diagonal at high scores and *above* it at low ones — its 0.9 is really 0.75, its 0.1 really 0.25. Lecture 03 showed a model that is over-confident by construction (naive Bayes counting correlated evidence twice). Only a calibrated score can go into Lecture 01's cost threshold p* = 0.1; a 0.9 that means 0.75 moves every decision near the cut.

What Maximum Likelihood Guarantees

At the peak, the gradient for the intercept ($x_0 = 1$) is zero:

$$\sum_{i=1}^{n}(y_i - h_i) = 0 \quad \Longrightarrow \quad \frac{1}{n}\sum_{i=1}^{n} h_i = \frac{1}{n}\sum_{i=1}^{n} y_i$$

What does this equation promise — and what does it not?

Speaker notes

Math first: the equation is Module 3's gradient, read for the intercept only. It is a genuine guarantee, but a weak one: it holds on average and on the data the model was fitted to. Per-score calibration needs the model to be right (correctly specified), which a linear log-odds rarely is exactly; in practice logistic regression is often close, and that is why it has a reputation for "calibrated probabilities" — a reputation, not a theorem. Separation (ranking fraud above legitimate cases) is a different property, measured in Lecture 06.

What Maximum Likelihood Guarantees

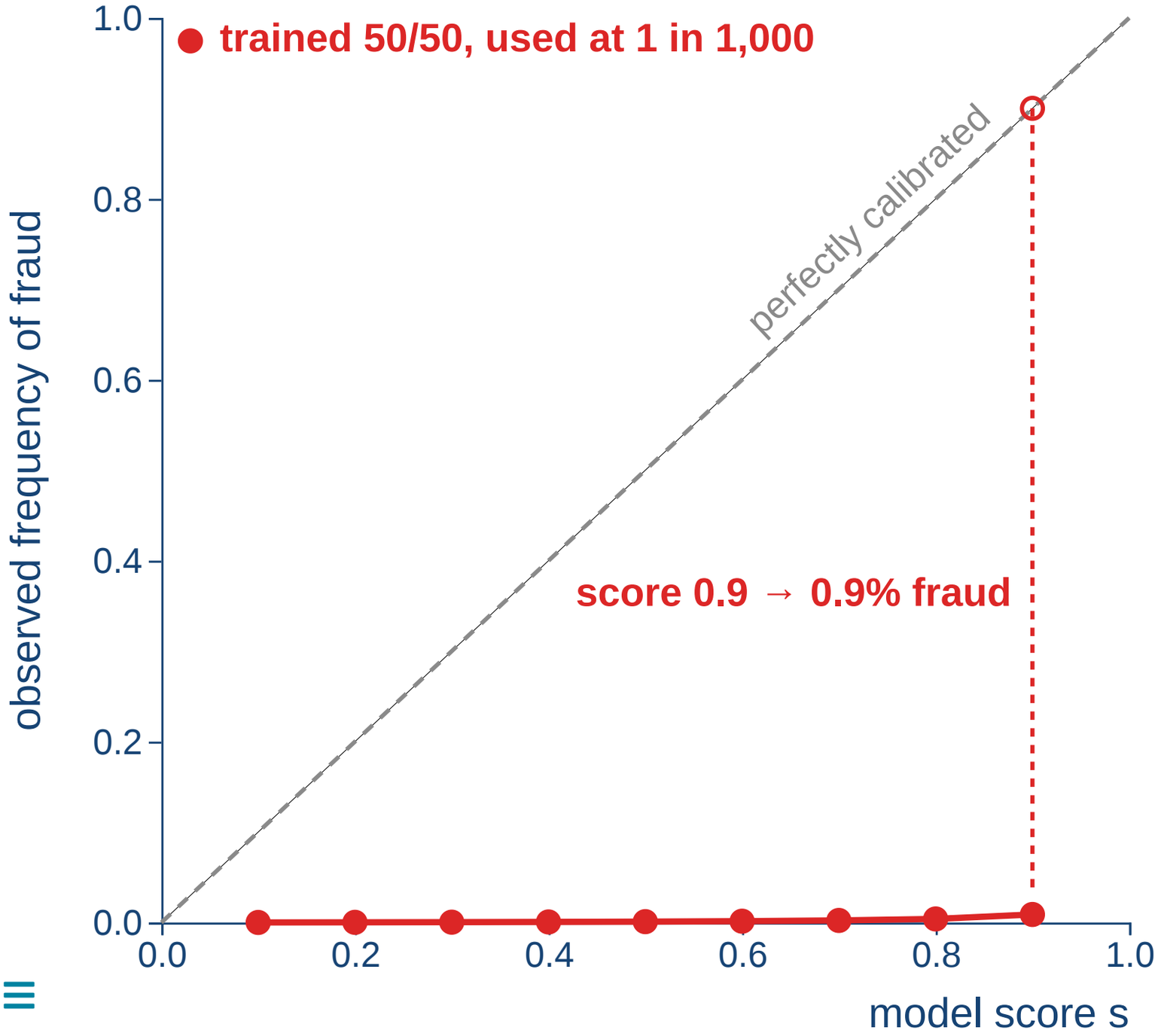
At the peak, the gradient for the intercept ($x_0 = 1$) is zero:

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What does this equation promise — and what does it not?

promised	not promised
average score = training fraud rate: *calibration in the large*	each score right — needs log-odds truly linear in $\mathbf{x}$
on the **training** distribution	after a base-rate shift, or **rebalancing**
	that it **separates**: the base rate for everyone is calibrated — and useless

Rebalanced Training



Fraud is 1 in 1,000, so practitioners **undersample** legitimate cases to 50/50. The model learns odds for a world where fraud is common.

Bayes in odds form:

$$\text{real odds} = \text{model odds} \times \frac{1/999}{1}$$

Score 0.9: odds $9 \to 9/999$, about **0.9%**.

Speaker notes

The click shows the rebalanced model on real traffic: every point hugs the floor. A score of 0.9 means about 0.9% fraud (odds 9/999, probability 9/1008 = 0.89%). The repair is Bayes' theorem from Lecture 02 in odds form: undersampling legitimate cases changes the prior, not p(x | y), so the model's odds are the likelihood ratio times the *training* prior odds; swap in the real prior odds. General form: real odds = model odds × (real prior odds) / (training prior odds); here (1/999) / 1. The base rate that made a "99% accurate" flag mean 9% in Lecture 02 returns — and a threshold of 0.5 on the raw score would block almost only honest customers.

Regularisation Is a Prior

$$\min_{\mathbf{w}} \mathrm{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$$

You know this as ridge regression or weight decay. Which prior's peak is it?



Speaker notes

Multiply the log-posterior by $-2\sigma^2$ to get the penalised objective; $\lambda = \sigma^2/\tau^2$ reads well: noisy data (large σ^2) or a tight prior (small τ^2) means more shrinkage. This is Lecture 02's Beta(2, 2) exercise one level up: the prior pulled the peak from 0.8 to 0.75, the Gaussian prior pulls weights towards 0. For logistic regression the same prior gives cross-entropy + $\|w\|^2/(2\tau^2)$ — the finite peak of the runaway slide. The intercept is usually left unpenalised, so calibration in the large survives. L1: the Laplace log-density $-|w|/b$ has a corner at 0, and at a corner the peak can sit exactly at 0 over a whole range of data; averaging over the posterior smooths the corner away.

Regularisation Is a Prior

$$\min_{\mathbf{w}} \mathrm{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$$

You know this as ridge regression or weight decay. Which prior's peak is it?

MAP from Lecture 03 with $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$: $\log p(\mathbf{w} \mid D) = -\frac{\mathrm{RSS}}{2\sigma^2} - \frac{\|\mathbf{w}\|^2}{2\tau^2} + \text{const}$, so $\lambda = \sigma^2/\tau^2$.

penalty	prior on $\mathbf{w}$	what its peak does
$\lambda \|\mathbf{w}\|^2$	Gaussian	shrinks; keeps separable data finite
$\lambda \|\mathbf{w}\|_1$	Laplace	exact zeros — a property of the **peak** (the kink at 0); the posterior mean has none

Mini-Exercise

A fraud model trained on a **50/50** sample is used on real traffic, where fraud is **1 in 1,000**. It outputs **0.5** and **0.8**.

What are the real fraud probabilities?



Speaker notes

One minute to read, one to think; the odds 1 and 4 are chosen so that the arithmetic is trivial. The punchline of the first answer: a coin-flip score on balanced data means "nothing unusual" on real traffic. Exact values: 1/1000 = 0.100% and 4/1003 = 0.399%.

Mini-Exercise

A fraud model trained on a **50/50** sample is used on real traffic, where fraud is **1 in 1,000**. It outputs **0.5** and **0.8**.

What are the real fraud probabilities?

Multiply the odds by $1/999$. Score 0.5: odds $1 \to 1/999$, about **0.1% —** the base rate itself. Score 0.8: odds $4 \to 4/999$, about **0.4%**.

Takeaway

A score is a probability when it is **calibrated** — checkable on a reliability diagram, not by accuracy. Maximum likelihood guarantees it only on average and only on the training distribution; **rebalancing** breaks it, and the prior odds repair it. **Regularisation is a prior**, and a penalised fit is a MAP estimate.

Every Loss So Far Is a Likelihood

So far — Lecture 06 meets losses that are not.

you minimised	it is $-\log$ of	the fit is
squared error	Gaussian noise around $\mathbf{w}^\top \mathbf{x}$	the peak (MLE)
cross-entropy	Bernoulli labels with bias $\sigma(\mathbf{w}^\top \mathbf{x})$	the peak (MLE)
$+\lambda \lVert \mathbf{w} \rVert^2$	a Gaussian prior on $\mathbf{w}$	a prior's peak (MAP)
$+\lambda \lVert \mathbf{w} \rVert_1$	a Laplace prior on $\mathbf{w}$	a prior's peak (MAP)

Dropped: the spread

how sure we are about $\mathbf{w}$ — **Lecture 07** keeps it

Dropped: $p(\mathbf{x})$

what a normal input looks like — **Lecture 03** modelled it, naive Bayes

Speaker notes

The lecture in one table; point at the argument map's takeaway box. Read each row as a translation: a loss is a noise model, a penalty is a prior, a fitted model is a peak. Say "so far" out loud: the hinge loss and the 0/1 error are not negative log-likelihoods, and Lecture 06's shortcut (keep the loss) is exactly what remains when there is no probability model behind a loss. The two cards point forward: the villa's band (the spread, Lecture 07) and the unusual transfer (p(x), Lecture 04).